Score for this attempt: **4** out of 4

Question 1

1 / 1 pts

We choose an integer uniformly at random between 1 and 602. What is the probability that the chosen number is divisible by 4?

$\#\{n \leq 602 : n \text{ divisible by } 4\} = 600/4 = 150$

(This is b/c every 4th integer is divisible by 4.)

Provide a numerical answer with precision up to 3 decimal places.

0.249

If we choose n uniformly at random, then

$P(n \text{ divisible by } 4) = \frac{150}{602} \approx 0.249$

Question 2

1 / 1 pts

We roll a die 2 times and then flip a coin 3 times. How many different outcomes can we have for this experiment where <u>there are no two die rolls with the same number and not all coin flips are tails?</u>

↳ call such an outcome 'successful.'

(Your answer should be an integer number.)

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A successful outcome consists of

$\underbrace{\text{An ordered pair of 2 distinct \#s in } \{1,\dots,6\}}_{(6\times 5)}$ AND $\underbrace{\text{a string of 3 coin flips, not all tails}}_{7}$

$\times$

Question 3

1 / 1 pts

Let $q = \sum_{k=1}^{\infty} \frac{1}{2^{ak+b}}$, where a =4 and b =3. Find the exact value of $\frac{1}{q}$.

(Hint: it will be a positive integer.)

$$q = \sum_{k=1}^{\infty} \frac{1}{2^{ak+b}} = \frac{1}{2^b} \sum_{k=1}^{\infty} \left(\frac{1}{2^a}\right)^k$$

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$$= \frac{1}{2^b} \cdot \frac{1}{2^a} \sum_{k=0}^{\infty} \left(\frac{1}{2^a}\right)^k = \frac{1}{2^b \cdot 2^a} \left(\frac{1}{1-\frac{1}{2^a}}\right)$$

$$= \frac{1}{8}\left(\frac{1}{16}\right)\left(\frac{1}{1-\frac{1}{16}}\right) = \frac{1}{8}\left(\frac{1}{16}\right)\left(\frac{16}{15}\right)$$

Question 4

1 / 1 pts

Consider an experiment with sample space $\Omega = \{a, b, c, d, e\}$. You are given the following information: the probability of the event $\{a\}$ is 0.24, the probability of the event $\{c, d\}$ is 0.29, and the probability of the event $\{e\}$ is equal to 0.30. What is the probability of the event $\{c, d, e\}$?

Provide a numerical answer with precision up to 3 decimal places.

Note: $\{c,d,e\} = \{c,d\} \cup \{e\}$

and $\{c,d\} \cap \{e\} = \emptyset$,

0.59

so $P(\{c,d,e\}) = P(\{c,d\}) + P(\{e\})$

$= 0.29 + 0.30 = 0.59$

Quiz Score: **4** out of 4